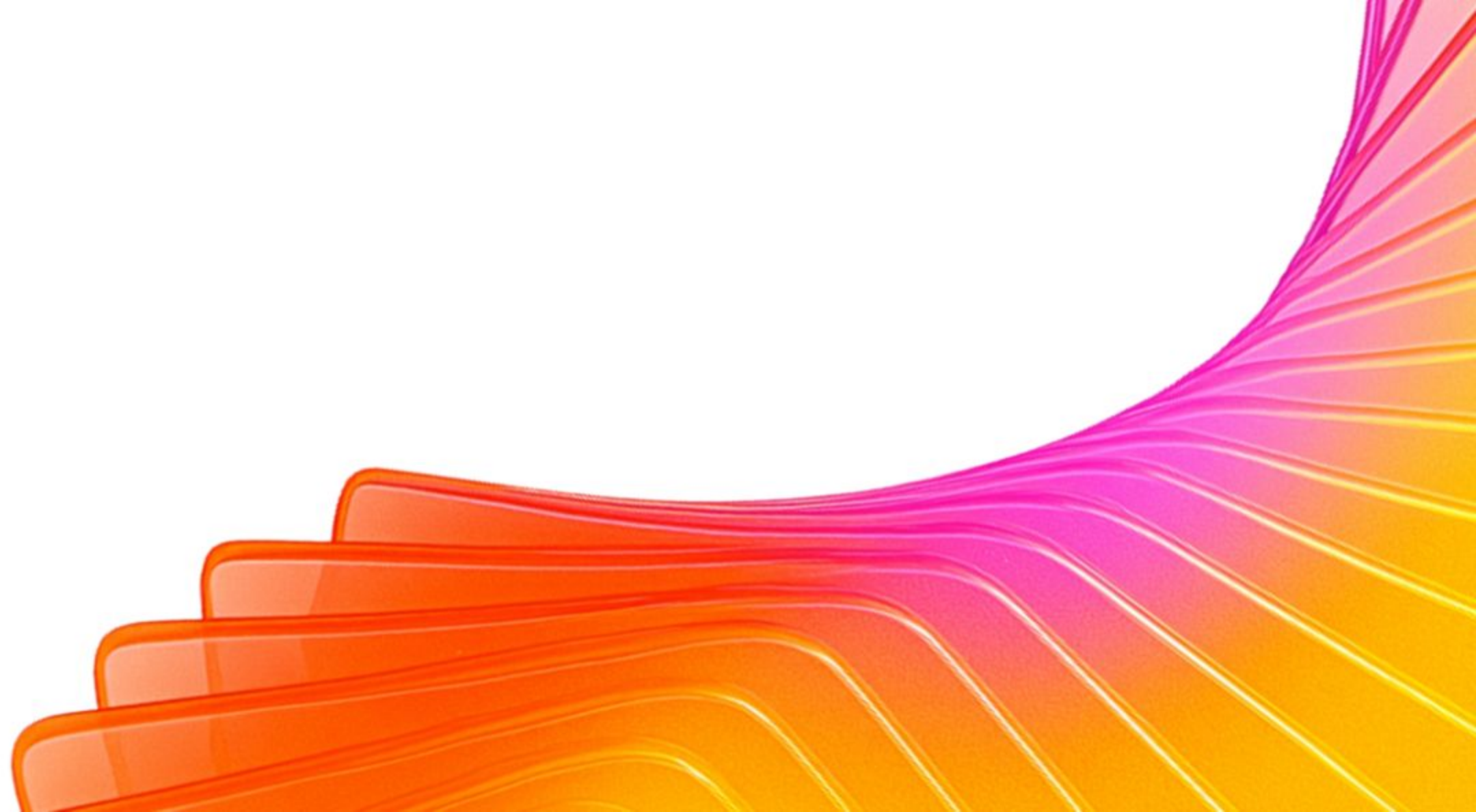


Gen AI Academy

APAC Edition

Build in APAC. Build for the world.



BuyOrWait

To Buy or Not to Buy — Steam sentiment intelligence

PARTICIPANTS

CuDecision

Team name

Wen Sihao — Team Leader

PROBLEM STATEMENT

AI-Powered Decision Intelligence Platform

Use case — Community Intelligence & Engagement: turning 114M Steam reviews (citizen feedback at platform scale) into personalised, auditable purchase decisions.

Two numbers from 114 million reviews

Steam's rating tells you if a game is good. Not if it's good for you.

SCORE ONE · THE GAME

Purchase Confidence

Playtime-weighted, 90-day half-life — is it good right now?

- Review-bombing detection

Rolling z-scores flag the episode; Gemini names the cause.

- Plain-English answers

Aggregates queried through Gemini NL→SQL.

SCORE TWO · YOU

Personal Fit

Weighed against your schedule, hardware and play style.

- 8 friction dimensions

Graded by Vector Search over negative reviews.

- Nightly sync

Steam API top-up anchors every score to today.

DATA SCALE

114.3M

Reviews re-scored

CORE OUTPUT

2 Scores

Per game, never blended

GPU LATENCY

2.75s

Full batch on an NVIDIA L4

COMPUTE SPEEDUP

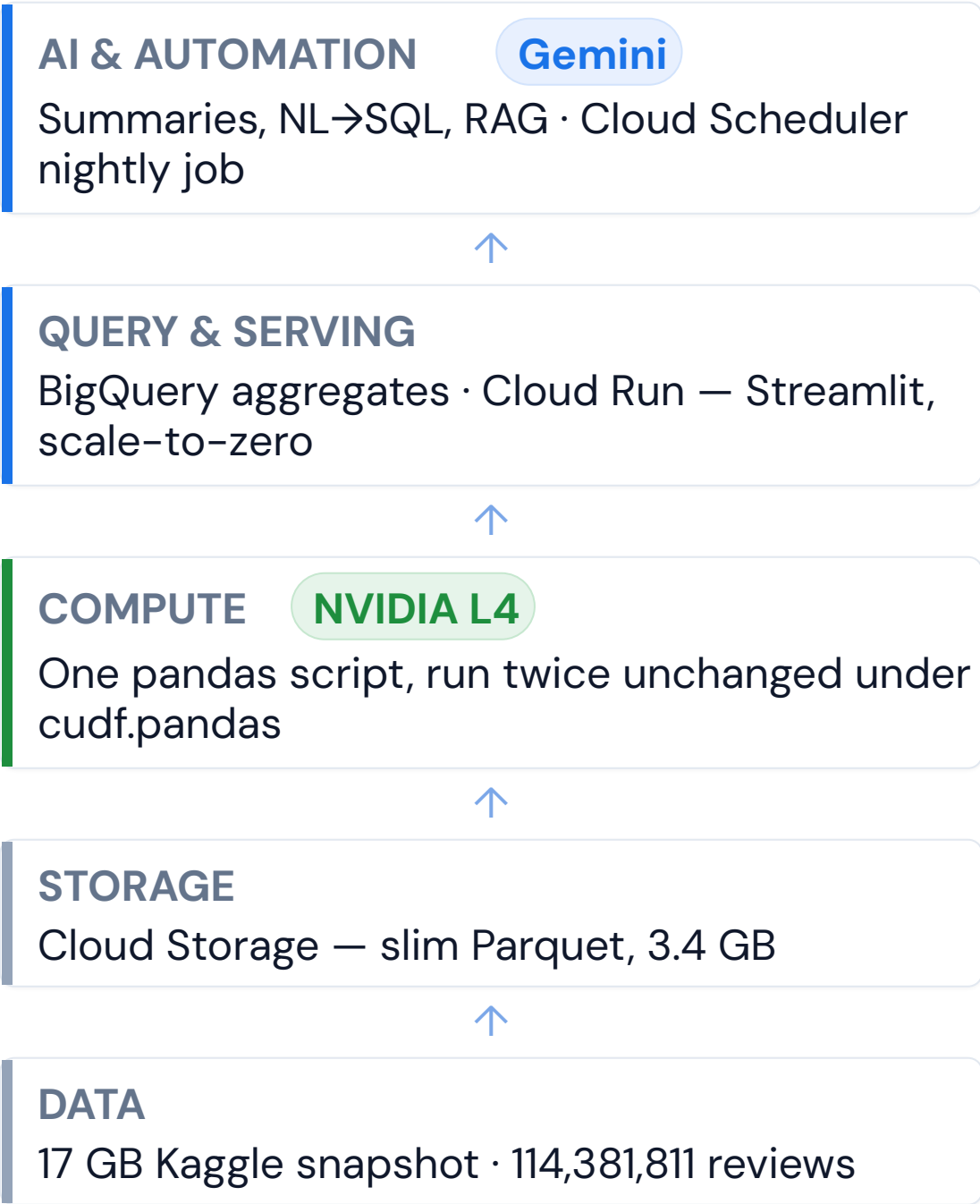
14x

Faster than 8 vCPUs

Approach, problem and core workflow

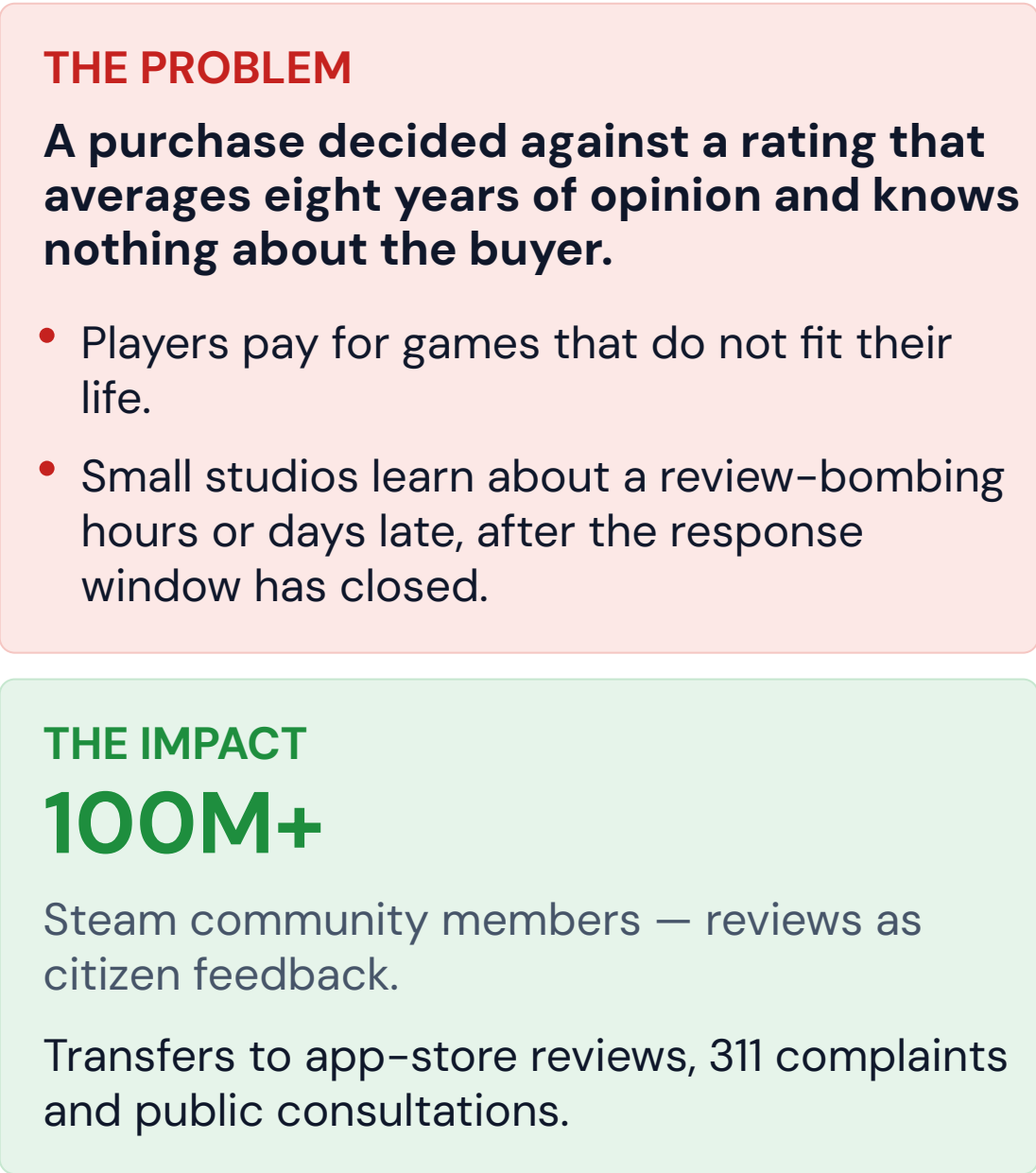
A · APPROACH

A 17 GB snapshot to a live verdict, RAPIDS on an NVIDIA L4.



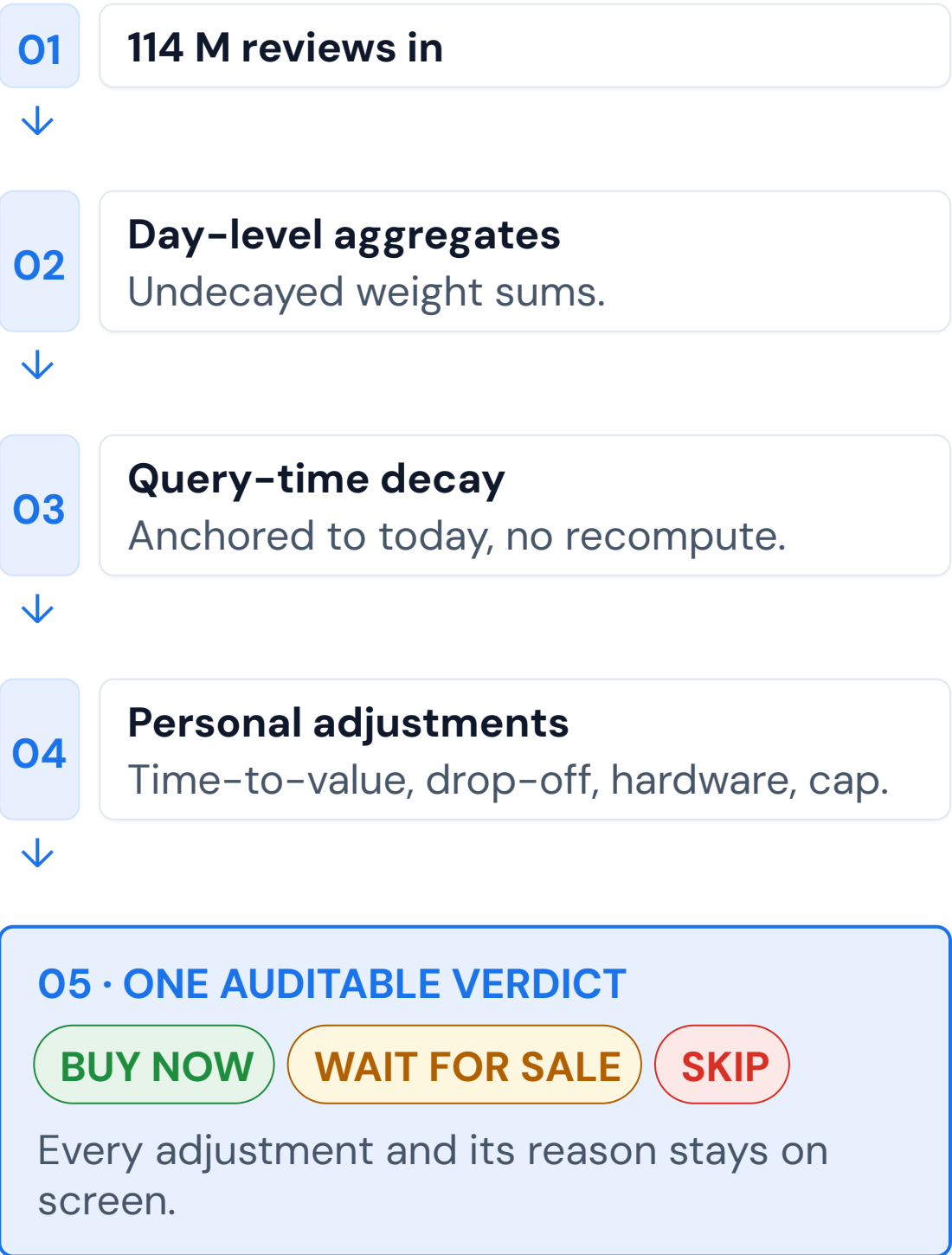
B · REAL-WORLD PROBLEM

Who today's rating fails, and who this pipeline serves next.



C · CORE WORKFLOW

Five steps from raw reviews to an auditable decision.



Differentiation and unique selling proposition

Key contrasts separating BuyOrWait from a traditional crowd rating.

01 Person–game relationship

ELSEWHERE

One crowd rating stands in for every buyer.

BUYORWAIT

Quality and fit stay separate numbers, so “great game, wrong game for you” remains visible.

02 Explains, not just detects

ELSEWHERE

Generic spike alerts with no context.

BUYORWAIT

Gemini root-cause attribution — PVE cancelled, monetisation, hero locking — against the game’s own 60-day baseline.

03 Evergreen by construction

ELSEWHERE

Scores are as stale as the last batch.

BUYORWAIT

Query-time exponential decay: the score moves daily with zero recomputation and the nightly delta only appends.

04 Percentile–relative friction grading

Graded against each dimension's own distribution across the corpus: some themes draw complaints in two thirds of all games and others in under 5 %, so one flat threshold would grade every game identically.

05 Anti-manipulation defence in the metric

$\log(1 + \text{playtime})$ weighting

Dual anomaly test — $z > 3$ and $2\times$ baseline

Free-key share per game

What the product does

Core capabilities and the reasoning engine behind them.

- PERSONALISED DECISION ENGINE

01 Personal Fit Score

Itemised adjustments, one verdict: Buy / Wait / Skip.

02 Purchase Confidence Score

Playtime-weighted, 90-day half-life, evergreen.

- FRICITION & BOMBING RADAR

03 Friction Radar

Eight dimensions graded against the whole corpus.

04 Review-Bombing Alerts

Z-score episodes with a Gemini-written cause.

- CONVERSATIONAL AI & GROUNDED RAG

05 Ask Gemini

Natural language to BigQuery SQL, guarded and cost-capped.

06 Ask the Reviews

RAG over **BigQuery VECTOR_SEARCH**, with citations.

- LIVE ANALYTICS & NIGHTLY SYNC

07 Player Ownership X-ray

Direct-paid, free-key and Early-Access mix per game.

08 Live Check + nightly sync

Today's Steam sentiment, even for post-snapshot releases.

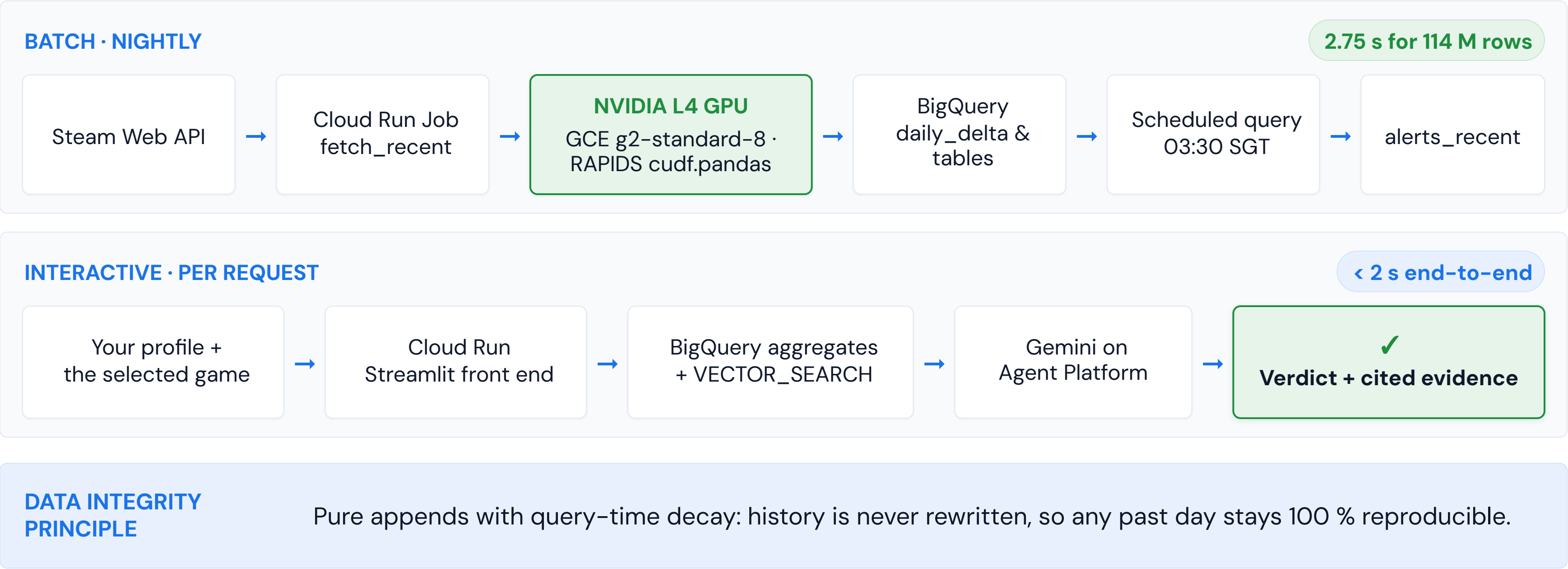
Live-queried guarantee — every number on every page is queried at request time, never hand-typed.

BUY NOW

WAIT FOR SALE

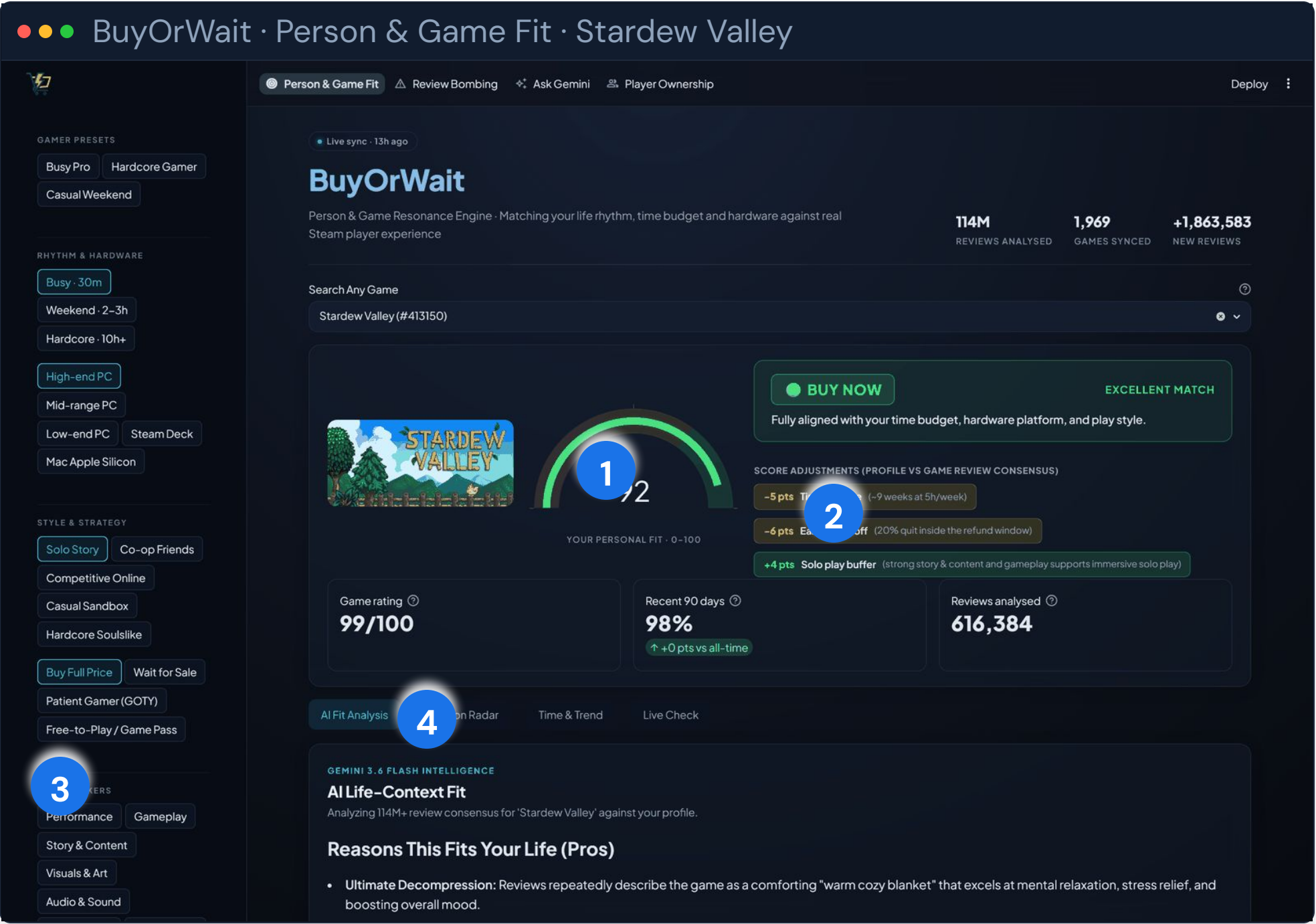
SKIP

Dual-pipeline system architecture



The Person & Game Fit screen

An interactive breakdown of the personalised decisioning UI.



1 Dual scoring architecture
Game quality against your life rhythm — 92/100 here.

2 Itemised fit adjustments
-5 time to value, -6 early drop-off, +4 solo buffer.

3 Dealbreaker hard cap
Any dealbreaker at High Risk caps the fit at 35.

4 Four deep-dive tabs
AI Fit · Friction Radar · Time & Trend · Live Check.



FRICION RADAR TAB
Eight thematic dimensions, each ranked against every indexed game.

End-to-end enterprise architecture on Google Cloud

ORCHESTRATION CONTROL PLANE



SOURCES

Kaggle snapshot
— 114 M reviews

Steam Web API —
nightly top-up

INGEST & COMPUTE

Cloud Storage — Parquet,
3.4 GB

NVIDIA L4 GPU
GCE g2-standard-8 ·
RAPIDS

FastEmbed CUDA
bge-small-en-v1.5 · 300 k

DATA · BIGQUERY

Tables
game_daily · game_scores · alerts ·
daily_delta

Vector index
review_vectors — 384-d IVF/COSINE

Views
v_daily_all · v_scores_live · v_freshness

AI & REASONING

Gemini 3.6 Flash
Cause summaries and
episode attribution.

Grounded RAG
Natural language to
SQL, with citations.

SERVE

Streamlit UI
Cloud Run —
scale-to-zero, 2
GiB.

Looker Studio
Dashboards and
analytics.

CROSS-CUTTING · PRODUCTION HARDENING

- **PERFORMANCE**
App cache 10 min → BQ result
cache → pre-aggregation.

- **SECURITY**
Keyless IAM service accounts —
zero JSON keys.

- **GUARDRAILS**
Read-only SQL guard, 1 GB scan
cap, AI rate limits.

- **OBSERVABILITY**
Cloud Monitoring uptime and
job-failure alerts.

Stack selection and trade-offs

Why this tech stack, what it costs to run, and how much headroom is left.

CATEGORISED TECH STACK

NVIDIA ACCELERATED COMPUTE

- RAPIDS cudf.pandas
- NVIDIA L4 on GCE g2
- RMM pool allocator
- CUDAExecutionProvider

GOOGLE CLOUD INFRASTRUCTURE

- BigQuery + VECTOR_SEARCH
- Gemini 3.6 Flash · Agent Platform
- Cloud Run — scale-to-zero
- Looker Studio
- Keyless IAM
- Cloud Build — CodeMender

WHY THIS STACK

14× faster

On one ~\$1/hr box

01 · RAPIDS over more CPUs

Iterative dataframe logic, not one SQL shape. Where SQL is the better tool — the nightly alert recompute — we use SQL.

\$0 idle

300 k vectors already in BigQuery

02 · BigQuery vectors over a managed DB

Every query filters to one game, so no new service and no idle cluster. Past a few million vectors that trade flips.

~\$0.20/day

~10 min of GPU a night

03 · Cloud Run over GKE

One stateless container; idle cost is effectively zero and the GPU box stays off unless a batch runs.

SCALABILITY

10× headroom at 2.75 s

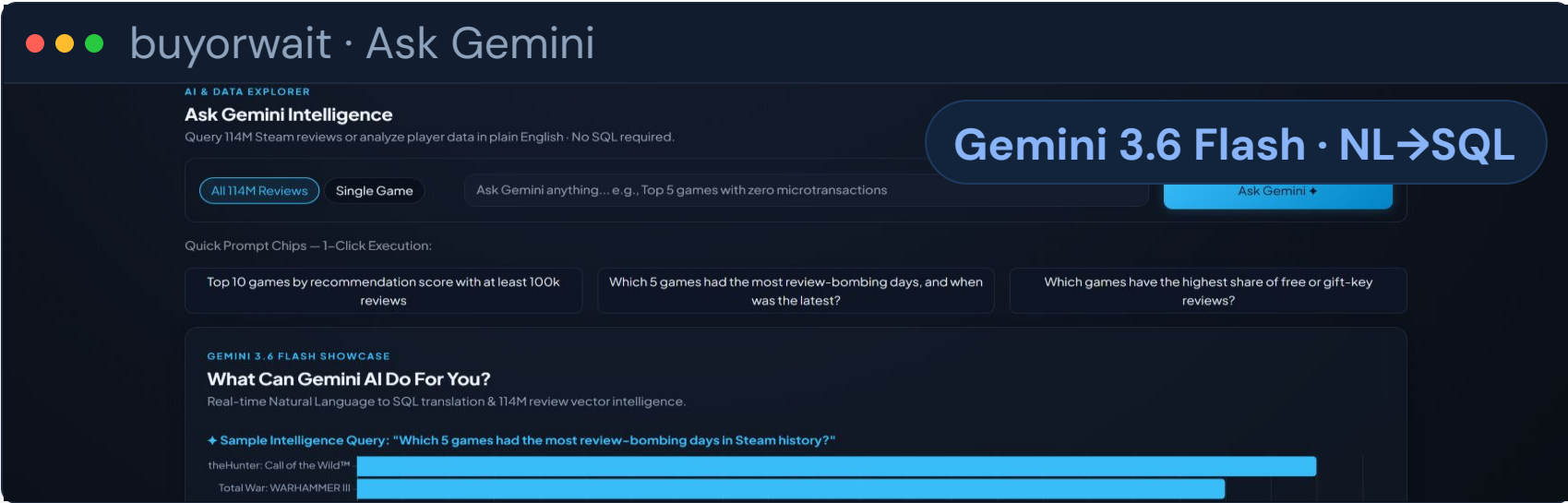
Scale-out path: Dataproc Serverless with Spark RAPIDS, and a managed vector store only once the corpus outgrows a per-game filter.

The prototype in use

Live dashboards, and the NVIDIA L4 run behind the 2.75 s batch.



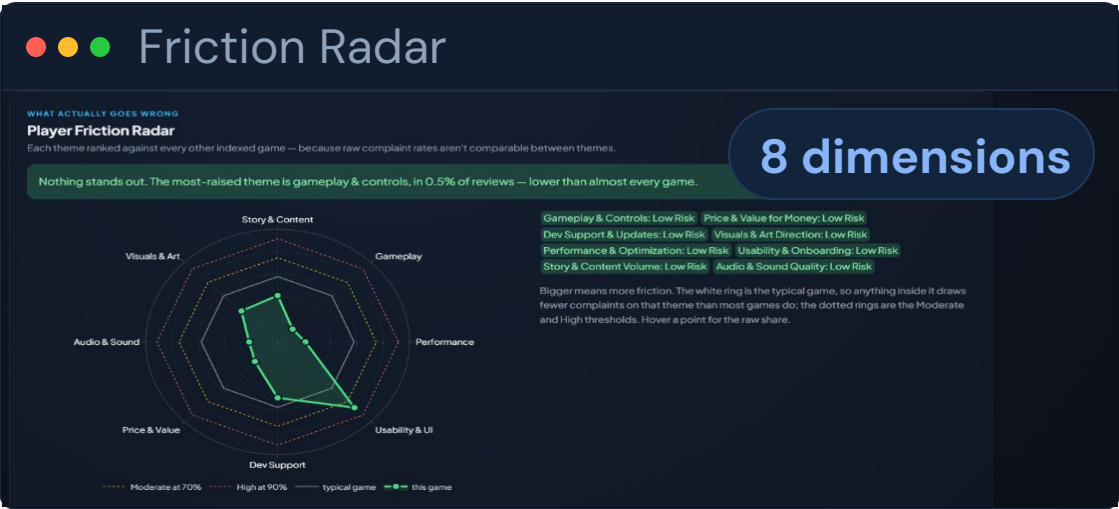
Review-bombing engine — episode detection with a Gemini root cause.



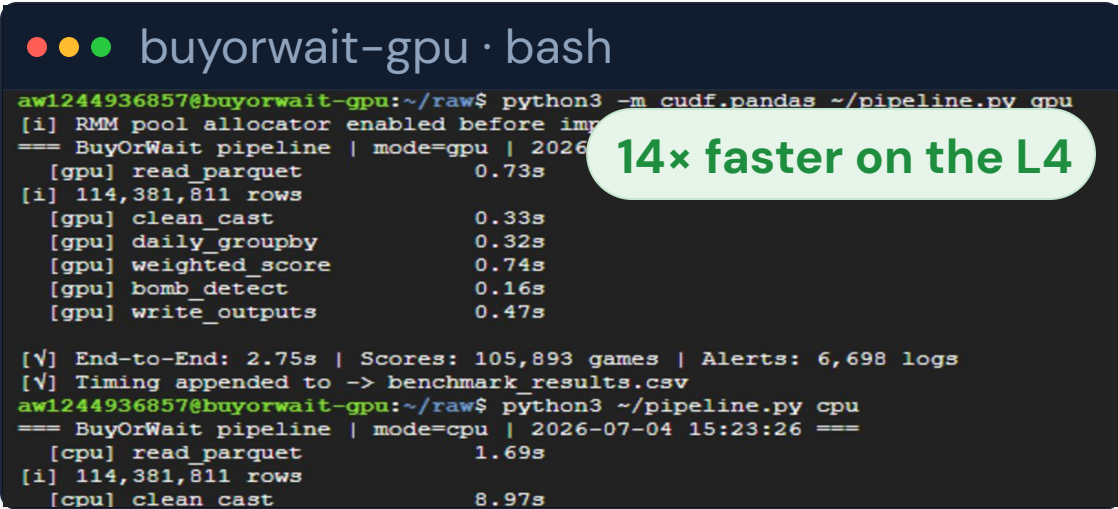
Ask Gemini — plain English over BigQuery aggregates, cost-capped.



Player Ownership X-ray — 64.6 % average direct-paid share.



Friction Radar — eight themes ranked against the whole corpus.



L4 batch run — 114,381,811 rows in 2.75 s versus 38.32 s on 8 vCPUs.

What changed in the refinement phase

Engineering velocity and production hardening, dimension by dimension.

DIMENSION	BEFORE · TOP-100 PHASE	AFTER · REFINEMENT PHASE
Data ingestion	☐ Static 2023 snapshot	✓ Evergreen scores + nightly Steam sync Cloud Scheduler Cloud Run Job
Scoring logic	☐ Crowd score only	✓ Personal Fit Score with itemised, auditable adjustments
AI attribution	☐ Keyword-free alerts	✓ Gemini cause attribution per bombing episode
Vector RAG	☐ No text retrieval	✓ ~300 k GPU-built embeddings + BigQuery VECTOR_SEARCH
Risk grading	☐ Flat risk thresholds	✓ Percentile-relative grading across the whole corpus
User experience	☐ Prototype UI	✓ Production UI — st.navigation, native theme, mobile pass
Security & guardrails	☐ No abuse or failure guards	✓ AI rate limits, read-only SQL guard idempotent MERGE, job alerts

Top-100 phase measured the game. This phase measures you and the game.



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Thank you

LIVE DEMO <https://buyorwait-1047454501331.asia-southeast1.run.app>

DEMO VIDEO https://drive.google.com/file/d/1aMJzKrKsfrs9s_pNlaYNO3KzuOqPdaSb/view?usp=sharing

SOURCE <https://github.com/hypoxic127/buyorwait>

TEAM CuDecision — Wen Sihao